

TransFOL: A Logical Query Model for Complex Relational Reasoning in Drug-Drug Interaction

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Abstract—Predicting drug-drug interaction (DDI) plays a crucial role in drug recommendation and discovery. However, wet lab methods are prohibitively expensive and time-consuming due to drug interactions. In recent years, deep learning methods have gained widespread use in drug reasoning. Although these methods have demonstrated effectiveness, they can only predict the interaction between a drug pair and do not contain any other information. However, DDI is greatly affected by various other biomedical factors (such as the dose of the drug). As a result, it is challenging to apply them to more complex and meaningful reasoning tasks. Therefore, this study regards DDI as a link prediction problem on knowledge graphs and proposes a DDI prediction model based on Cross-Transformer and Graph Convolutional Networks (GCNs) in first-order logical query form, TransFOL. In the model, a biomedical query graph is first built to learn the embedding representation. Subsequently, an enhancement module is designed to aggregate the semantics of entities and relations. Cross-Transformer is used for encoding to obtain semantic information between nodes, and GCN is used to gather neighbour information further and predict inference results. To evaluate the performance of TransFOL on common DDI tasks, we conduct experiments on two benchmark datasets. The experimental results indicate that our model outperforms state-of-the-art methods on traditional DDI tasks. Additionally, we introduce different biomedical information in the other two experiments to make the settings more realistic. Experimental results verify the strong drug reasoning ability and generalization of TransFOL in complex settings.

Index Terms—Drug-drug interaction, logical query, transformer, graph convolutional network, knowledge graph.

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I. INTRODUCTION

DRUG-DRUG interaction (DDI) prediction is critical for drug repositioning, drug recommendation, and drug discovery, and unknown changes in drug function can result in adverse drug reactions [1], [2]. Predicting potential interactions between factors such as drugs, diseases, and proteins is essential for avoiding severe side effects. While adverse reactions can be found in clinical trials, some serious ones are only discovered when patients receive treatments. Detection and prevention of adverse reactions require significant workforce and financial resources, and achieving a safe and reliable clinical trial is challenging. Therefore, automatic analysis methods have been proposed to address this issue [3], [4], [5]. However, traditional machine learning models often require manual feature engineering, which requires professional knowledge and is time-consuming and labour-intensive.

Recent studies investigated Bayesian methods and deep learning models. For example, Takeda et al. [6] considered pharmacokinetics (PK) or pharmacodynamics (PD) side effects in DDI prediction and developed a Bayesian network [7] that combines molecular similarity and drug side effect similarity for drug effect prediction. CNN-DDI [5] uses convolution neural networks to learn feature representation for DDI prediction. However, these methods only consider drug-drug, drug-target, and drug-phenotype interactions as external knowledge, which may need to be more comprehensive for complex biomedical tasks. Knowledge graphs (KG) and knowledge representation learning methods [8] also facilitate the domain application of DDI prediction [9], [10], [11]. MCFF-MTDDI [12] solves feature redundancy and KG noise by effectively fusing drug chemical structure features, additional label features of drug pairs, and KG features of drugs, as well as using a multi-channel feature fusion module using gated recurrent units.

With the emergence of biomedical knowledge graphs, more types of entities and relationships are applied to this task. TriVec model [11] is a KG embedding model that projects entities and their relations to dense vectors for prediction through a neural network. STNN-DDI [13] proposes a substructure perception tensor model that maps drugs to substructure space with the capacity to perform multi-type DDI prediction in a unified form and an interpretable manner. However, most of them need to utilize the neighbourhood information of the target entity directly during reasoning, making the model unable to utilize the practical external knowledge fully. More importantly, they

can only predict interactions between the simplest pair of drugs and cannot be extended to more complex scenarios, which limits their application in the real world.

In such a context, this study frames the DDI problem as a link prediction task on a knowledge graph. It proposes a DDI prediction model named TransFOL, based on GCN and Cross-Transformer, to address the unique complex relational reasoning requirements within a **First-Order Logic** query framework. Relational reasoning refers to inferring potential interactions or dependencies between entities by analyzing their associations and attribute information. Complex relational reasoning in the pharmaceutical domain specifically denotes the reasoning tasks that necessitate considering multiple biomedical factors (such as drug dosage, metabolic pathways and disease background) or being able to handle multi-hop causal chains, conditional constraints, and non-linear effects. This complexity arises from the fact that DDI is influenced not only by the direct pairing of drugs but is also comprehensively regulated by other biomedical factors. TransFOL aims to solve complex reasoning problems that traditional methods struggle to handle effectively, thereby improving the accuracy and practicality of DDI prediction.

In the model architecture of TransFOL, triple transformations transform the knowledge graph into a directed query graph without edge attributes. We further enhance the semantic representations of entities and relations using the plug-and-play bidirectional aggregation functionality. Then, we employ a Cross-Transformer and a GCN decoder to collect neighbourhood information and predict drug-drug interactions.

From a specific task perspective, we comprehensively tested TransFOL's reasoning capabilities under various complex conditions on four tasks. In the multi-class task (Task 1), we predict the only interaction type between two drugs; in the multi-label task (Task 2), we simultaneously predict multiple interaction types.

In addition to the common simple DDI prediction tasks mentioned above, we investigated complex prediction with additional dose attribute information. For instance, in Task 3, we introduce drug dose information into DDI prediction. Such prediction with attribute information is more accurate and realistic, which is challenging for existing prediction methods. This demonstrates that TransFOL has a more robust drug knowledge discovery ability in complex settings.

Moreover, we design a more complex and valuable task (Task 4) that requires the model to infer human clinical symptoms based solely on disease information and mammalian clinical symptoms, which helps address the current disadvantage of excessively long clinical cycles in the drug development process, demonstrating the generalization of TransFOL.

In summary, this work introduces the TransFOL model, a novel approach to drug knowledge discovery through logical query-based reasoning. TransFOL stands out in its ability to express intricate interactions between biomedical entities through logical queries. It performs superior on four benchmark datasets across multiple tasks and outperforms several strong baselines.

II. PRELIMINARIES

Let $\mathcal{G} = (\mathcal{E}, \mathcal{R})$ denotes a KG, where $e \in \mathcal{E}$ denotes an entity and $r \in \mathcal{R}$ is a binary predicate (or relation) $r : \mathcal{E} \times \mathcal{E} \rightarrow \{\text{True}, \text{False}\}$ that indicates whether a relation holds for a pair of entities. Given the first-order logical existential ($\exists$) and conjunctive ($\wedge$) operations, the conjunctive queries are defined as:

$$\mathcal{Q}[a] \doteq ?A : \exists E_1, \dots, E_m, e_1 \wedge \dots \wedge e_n$$

where $e_i = r(c, E)$, with $E \in \{A, E_1, \dots, E_m\}$, $c \in \mathcal{E}$, $r \in \mathcal{R}$

or $e_i = r(E, E')$, with $E, E' \in \{A, E_1, \dots, E_m\}$, $E \neq E'$, $r \in \mathcal{R}$ (1)

where A refers to the (unknown) target entity of the query, $E_1, \dots, E_m$ refers to existentially quantified bound variables (i.e., unknown intermediate entity sets), and c refers to the source entity. Given the query $\mathcal{Q}$, the goal is to find its target entity set $\mathcal{A} \subseteq \mathcal{E}$ that satisfies $a \in \mathcal{A}$ if and only if $\mathcal{Q}[a]$ is true.

Here, we use a simple example to explain how logical queries work. To convert the question - “*What are the diseases that opioid analgesics can treat?*” - into a logical query, we need to perform 2-hop reasoning under certain constraints, i.e., to follow a series of steps (in plain language) to obtain the query answer. First, we search for an opioid drug entity, i.e., the first constraint. Then, we look for a drug entity that can relieve pain, i.e., the second constraint that is the same as the previous one. Finally, we find the disease treated by the obtained entity in the second hop reasoning. This logical query is a common “2p”-type query that uses 2-hop knowledge graph queries. By analyzing the logical query, we can see that the second hop of the query provides the answer we need, which is the diseases treated by the obtained entity.

In addition to conjunction queries ($\wedge$), FOL queries include disjunction queries ($\vee$). Based on the knowledge of *Discrete Mathematics*, we know that a disjunctive standard form can be decomposed into several conjunctive typical forms, enabling the application of a rule to synthesize the conjunctive results of disjunctive predictions. In other words, all positive first-order logical queries can be transformed into typical disjunctive forms for unified processing.

III. METHODS

This section introduces TransFOL, a first-order logical query model based on Cross-Transformer and graph convolutional network (GCN) for solving DDI prediction problems on biomedical knowledge graphs. To enable TransFOL to aggregate complex semantics and neighbour information effectively, we propose a Cross-Transformer as an encoder and employ an aggregation function enhancement module and a GCN decoder before and after encoding as the further enhancement.

In Section III-A, we summarize the overall structure of TransFOL. Section III-B describes the triplet transformation strategy that enables heterogeneous graphs to be transformed into directed logical query graphs without edge attributes. Section III-C introduces a plug-and-play representation enhancement module.

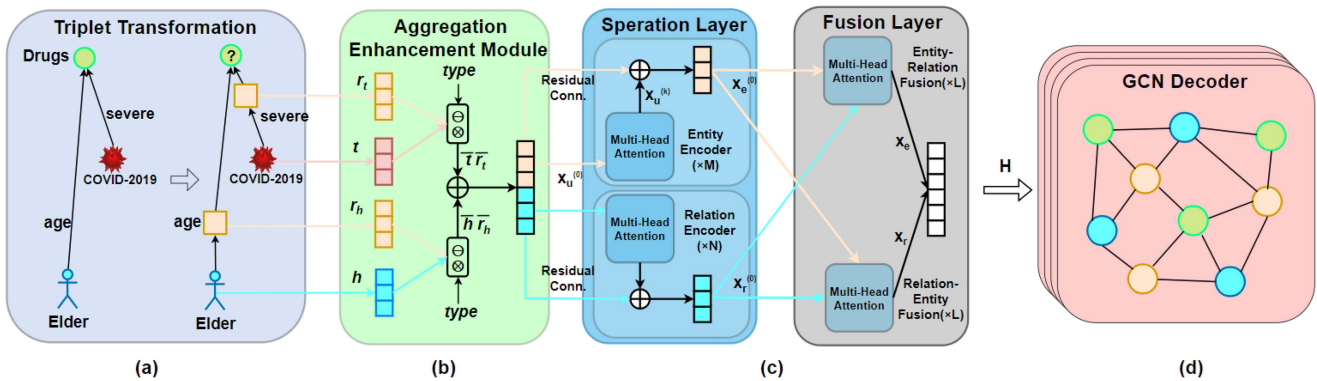
Example:**What drugs can cure the elderly with serious COVID-19?**

Fig. 1. Overall structure of the TransFOL model. **(a)** The triplet transformation module treats the attribute information of entities as relationships in a logical query graph. It converts them into node form, enabling the unified processing of sparse relationships between medical entities. **(b)** The aggregation enhancement module is designed to aggregate semantic information. The pairwise representation integration part utilizes type information to integrate the head entity node h and head relationship node r_h , as well as the tail entity node t and tail relationship node r_t . Then, the semantic transfer part combines the head and tail information. It employs a plug-and-play bidirectional attention mechanism to integrate the representations of entities and relationships, thus obtaining enhanced representations. **(c)** The Cross-Transformer consists of two layers: the feature separation layer and the fusion layer. In the feature separation layer, the model encodes the entity X_e and the relationship X_r separately. They are fused in the feature fusion layer, and the model learns global features through flattening. This approach allows the model to capture rich entity and relationship features while achieving semantic propagation across entity types through specific cross-attention mechanisms. **(d)** GCN extracts valuable information from multi-hop neighbours through convolutional operations and then aggregates local neighbourhood information.

Section III-D presents a fusion model that uses the Cross-Transformer and GCN as the backbone neural network and leverages the structure of the knowledge graph and drug-related knowledge information to learn drug representations further.

A. Overall Structure of TransFOL

The framework of TransFOL is shown in Fig. 1. Our model comprises four modules: triplet transformation, representation enhancement module, Cross-Transformer, and GCN decoder.

First, to enable unified modelling of entities and relations, we adopt a triple transformation operation, converting relation edges into relation nodes to transform the KG into a directed graph without edge attributes. We convert knowledge graph triples into a kind of multi-entity tuples in the triple transformation module. Specifically, we transform relations into relation nodes and convert the knowledge graph into a directed graph without edge attributes. We employ this approach to transform the entity relationships in the knowledge graph into a logic query form and convert the edge attribute information into node representations, enabling the DDI prediction task to be handled in a unified node embedding space while leveraging the typically sparse nature of biomedical entity relationships to ensure computational efficiency.

After the triple transformation operation, we convert the knowledge graph into a heterogeneous query graph. We then use a representation enhancement module to aggregate semantic information. We employ a plug-and-play bidirectional attention mechanism to integrate the representations of entities, relations, and types, enabling the intermediate variable nodes to perceive the information of anchor nodes or target nodes and the relevant information in the reasoning chain, avoiding the interference

of redundant or irrelevant information. This helps to prevent weakening the connection between the anchor node and the target entity due to long-chain reasoning, allowing the model to understand better and capture the interactions between entities and their associations.

Next, we design a two-layer strategy in the Cross-Transformer module, including drug feature separation and fusion layers. In the drug feature separation layer, we encode entities and relations separately. In the drug feature fusion layer, we fuse the features and then use a flattening operation to learn global features. This enables the model to capture richer entity and relation features while achieving cross-entity type semantic propagation through a specific cross-attention mechanism.

Finally, in the GCN module, the model extracts valuable information from multi-hop neighbours through convolutional operations, thereby aggregating local neighbour information.

The model adopts a two-layer architecture of Cross-Transformer and GCN. This design effectively separates and integrates different features to learn global context, enhancing the model's ability to infer complex drug-drug interactions.

B. Triplet Transformation

We follow Liu et al. [14] to convert knowledge graph triplets into other multi-entity tuples in the triplet transformation module. For instance, in a DDI task, we convert triplets to quintuples: $(drug_1, r, drug_2) \rightarrow ((drug_1, r_1), r, (drug_2, r_2))$, as shown in Fig. 1. Specifically, we regard attribute information of entities as relationships in logical query graphs and transform these relationships into relational nodes, transforming the KG into a directed graph without edge attributes. In practice, triplet transformations are computationally inexpensive, as relational

graphs between medical entities such as drugs are typically very sparse.

Through triplet transformation, we convert the entity and relationship information in the heterogeneous graph into nodes in the query graph. The unified node embedding representation enables us to perform the same processing on entities and relationships in GCN.

We also add a particular node type embedding to distinguish between entity node embedding and relationship node embedding, which is beneficial for our subsequent separate operations on entities and relationships.

C. Representation Enhancement Module

1) *Pairwise Representation Integration*: Integrating the information of entities, relations, and node types can help to smooth the decision boundary when embedding queries. However, this requires capturing the target matching of queries in the KG. For instance, for the query “What drugs can cure the elderly with serious Covid-2019?”, we need to focus on the connection of “cure” in “COVID-2019” rather than the connection of “spread.” Similarly, we should only consider “cure” connections from “COVID-2019” and not “flu” or “cold.” To properly capture this constraint in the triplet, we introduce a bidirectional attention mechanism to integrate each state of a representation pair (“entity-relation”, “entity-type” and “relation-type”). Here, the two-way integrated representation between entity “Elder” and relationship “age” can be computed as:

$$\mathbf{M}_h^{\text{er}} = \text{ReLU}(\mathbf{W}_1[\mathbf{h} \ominus \mathbf{r}_h \parallel \mathbf{h} \otimes \mathbf{r}_h] + \mathbf{b}_1) \quad (2)$$

$$\mathbf{M}_h^{\text{re}} = \text{ReLU}(\mathbf{W}_2[\mathbf{r}_h \ominus \mathbf{h} \parallel \mathbf{r}_h \otimes \mathbf{h}] + \mathbf{b}_2) \quad (3)$$

where $\mathbf{h} \in \mathbb{R}^{d \times 1}$ is the embedding of entity node “Elder”, $\mathbf{r}_h \in \mathbb{R}^{d \times 1}$ is the embedding of relationship node “age”, $[\parallel]$ is the concatenation operation, $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{2^d \times 2^d}$ and $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^{2^d \times 1}$ are learnable parameters. We employ element-wise subtraction $\ominus$ and element-wise multiplication $\otimes$ to construct matching representations. $\mathbf{M}_h^{\text{er}} \in \mathbb{R}^{2^d \times 1}$ is the output of integrating relational information into entities, and $\mathbf{M}_h^{\text{re}} \in \mathbb{R}^{2^d \times 1}$ is the output of integrating entity information into relationships. Similarly, we can get the output of integrating type information into entities $\mathbf{M}_h^{\text{es}} \in \mathbb{R}^{2^d \times 1}$ and the output of integrating type information into relationships $\mathbf{M}_h^{\text{rs}} \in \mathbb{R}^{2^d \times 1}$ (where the type information is extracted from the dataset, in the same way, as [15]).

Through the bidirectional fusion of entities and relations, we simultaneously obtain a relation-aware entity representation $\mathbf{M}_h^{\text{er}}$ and an entity-aware relation representation $\mathbf{M}_h^{\text{re}}$, capturing entities and their relational interactions.

Next, we use a gating mechanism to combine the results of bidirectional fusion, as it better regulates the information flow. Taking the entity fusion representation as an example, the final representation of the entity using the relationship-aware entity is computed as follows $\mathbf{M}^{\text{er}}$ and the type-aware entity $\mathbf{M}_h^{\text{es}}$ representations as input:

$$\mathbf{G} = \text{ReLU}(\mathbf{W}_3 \mathbf{M}^{\text{er}} + \mathbf{W}_4 \mathbf{M}_h^{\text{es}} + \mathbf{b}_3 + \mathbf{b}_4) \quad (4)$$

$$\widetilde{\mathbf{M}}_h^{\text{e}} = \mathbf{G} \cdot \mathbf{M}_h^{\text{er}} + (1 - \mathbf{G}) \cdot \mathbf{M}_h^{\text{es}} \quad (5)$$

where $\mathbf{W}_3, \mathbf{W}_4 \in \mathbb{R}^{2^d \times 2^d}$ and $\mathbf{b}_3, \mathbf{b}_4 \in \mathbb{R}^{2^d \times 1}$ is learnable parameters. $\mathbf{G} \in \mathbb{R}^{2^d \times 1}$ is the reset gate. “ $\cdot$ ” represents the Hadamard product. $\widetilde{\mathbf{M}}^{\text{e}} \in \mathbb{R}^{2^d \times 1}$ is the final entity representation. We use a linear layer to project the fused features to the original vector size:

$$\bar{\mathbf{h}} = \mathbf{W}_5 \widetilde{\mathbf{M}}_h^{\text{e}} + \mathbf{b}_5 \quad (6)$$

where $\mathbf{W}_5 \in \mathbb{R}^{d \times 2^d}$ and $\mathbf{b}_5 \in \mathbb{R}^{d \times 1}$ is learnable parameters. $\bar{\mathbf{h}} \in \mathbb{R}^{d \times 1}$ is the final entity representation augmented by relations and types.

2) *Semantic Transfer*: The pairwise representation integration module primarily integrates entities and relations but needs to learn the complex semantics between drug entities and drug entities or between drug entities and interactions. Therefore, we utilize a set of specialized transfer functions to express the semantics between them explicitly:

$$\mathbf{h}' = \frac{1}{2} \text{Sin}((\mathbf{r}_h + \mathbf{r}_t) \cdot \mathbf{t}) \cdot \mathbf{h} \quad (7)$$

$$\mathbf{r}_h' = \text{Sin}(\mathbf{h} \cdot \mathbf{r}_t) \cdot \mathbf{r}_h \quad (8)$$

$$\mathbf{r}_t' = \text{Sin}(\mathbf{t} \cdot \mathbf{r}_h) \cdot \mathbf{r}_t \quad (9)$$

$$\mathbf{t}' = \frac{1}{2} \text{Sin}(\mathbf{h} \cdot (\mathbf{r}_h + \mathbf{r}_t)) \cdot \mathbf{t} \quad (10)$$

where $\mathbf{t}$ is the embedding of entity node, e.g., “Covid-2019”, $\mathbf{r}_h$ and $\mathbf{r}_t$ denote the relationship information of $\mathbf{h}$ and $\mathbf{t}$, respectively.

Since the number of relations is far less than the number of entity types in the biomedical knowledge graph, overly complex enhancement functions may lead to overfitting in relational modelling. To avoid this issue, entity and relation enhancement functions must have different forms when passed.

D. Cross-Transformer & GCN Decoder

We design a two-layer strategy comprising the Cross-Transformer module’s drug feature separation and fusion layers. In the separation layer, entities and relations are encoded separately. In the fusion layer, we fuse these two types of features and use the flatten operation to learn global features. In the GCN module, we aggregate the features learned in the above modules, learn local features through convolution functions, and further gather multi-hop neighbour information.

1) *Drug Feature Separation Layer*: After representing relationship edges as nodes, entity and relationship nodes are encoded separately using the same method. Notably, the number of relationship categories in bioinformatics is typically much less than that of entity categories. Therefore, complex relational encoding may perform poorly, and relational encoders usually have fewer layers than entity encoders. Here, we demonstrate how to encode entity nodes.

Given an embedding $\mathbf{x}_u^{(0)} \in \mathbb{R}^d$ for an entity node u , in the k -th drug feature separation layer, $\mathbf{d}_h = \frac{\mathbf{d}}{h}$, where h represents

the number of attention heads. The calculation formula for multi-head attention is:

$$\text{Attention}_i^{(k)} = \text{softmax} \left(\frac{\mathbf{Q}^\top \mathbf{K}}{\sqrt{d_h}} \right) \mathbf{V} \quad (11)$$

where $\mathbf{Q} = \mathbf{x}_u^{(k-1)} \mathbf{W}_Q$ and $\{\mathbf{K}, \mathbf{V}\} = \|\mathbf{x}_n \in \mathcal{N}(u) \mathbf{x}_n^{(k-1)}\{\mathbf{W}_K, \mathbf{W}_V\}, \mathbf{W}_{\{Q,K,V\}} \in \mathbb{R}^{d \times d_h}$, $\|$ denotes concatenation, $\mathcal{N}_u$ refers to the set of neighbors of node $\mathcal{N}(u)$.

Next, we apply a feed-forward neural network $\text{FFN}(\mathbf{x})$ on the output of attention weights projected by $\mathbf{W}_O \in \mathbb{R}^{d \times d}$:

$$\mathbf{x}_u^{(k)} = \text{FFN}(\|\mathbf{H}_{i=1} \text{Attention}_i^{(k)} \cdot \mathbf{W}_O) \quad (12)$$

$$\text{FFN}(\mathbf{x}) = \text{ReLU}(\mathbf{x} \mathbf{W}_6 + \mathbf{b}_6) \mathbf{W}_7 + \mathbf{b}_7 \quad (13)$$

where $\mathbf{W}_6, \mathbf{W}_7 \in \mathbb{R}^{d \times d}$ and $\mathbf{b}_6, \mathbf{b}_7 \in \mathbb{R}^d$ are learnable parameters. FNNs do not change the matrix size but can be useful for capturing patterns in the data. FNNs are equivalent to key-value networks used for memory- and attention-based models [16].

2) Drug Feature Fusion Layer: As shown in Fig. 1, after we get the output of the drug feature separation layer, we use two fusion encoders to aggregate entities and relations and then learn the global features of drugs through flattened and fully connected layers. Here, we show how to aggregate entities and relationships:

Given a drug feature separation layer entity encoding output embedding $\mathbf{x}_e^{(0)} \in \mathbb{R}^d$ and a drug feature separation layer relation encoding output $\mathbf{x}_r^{(0)} \in \mathbb{R}^d$, in the k -th Drug feature separation layer, $d_h = \frac{d}{h}$, where h represents the number of attention heads. The calculation formula for multi-head attention is:

$$\text{Attention}_i^{(k)} = \text{softmax} \left(\frac{\mathbf{Q}^\top \mathbf{K}}{\sqrt{d_h}} \right) \mathbf{V} \quad (14)$$

where $\mathbf{Q} = \mathbf{x}_e^{(k-1)} \mathbf{W}_Q$, $\{\mathbf{K}, \mathbf{V}\} = \mathbf{x}_r^{(k-1)} \mathbf{W}_{\{K,V\}} \in \mathbb{R}^{d \times d_h}$. Next, as with the drug feature separation layer, we apply a feed-forward neural network $\text{FFN}(\mathbf{x})$ on the output of attention weights projected by $\mathbf{W}_O \in \mathbb{R}^{d \times d}$:

$$\mathbf{x}_e^{(k)} = \text{FFN}(\|\mathbf{H}_{i=1} \text{Attention}_i^{(k)} \cdot \mathbf{W}_O) \quad (15)$$

$$\text{FFN}(\mathbf{x}) = \text{ReLU}(\mathbf{x} \mathbf{W}_8 + \mathbf{b}_8) \mathbf{W}_9 + \mathbf{b}_9 \quad (16)$$

where $\mathbf{W}_8, \mathbf{W}_9 \in \mathbb{R}^{d \times d}$ and $\mathbf{b}_8, \mathbf{b}_9 \in \mathbb{R}^d$ is learnable parameters. In contrast to the drug feature separation layer, the number of layers of the two fusion encoders should be equal since entities and relations are fused and aggregated in the fusion layers and are of equal standing. Finally, the fusion output $\mathbf{x}_e$ and $\mathbf{x}_r$ are flattened to learn global features:

$$\mathcal{H} = \mathbf{W}_{11}(\text{Flatten}[\mathbf{x}_e, \mathbf{x}_r] \mathbf{W}_{10} + \mathbf{b}_{10}) + \mathbf{b}_{11} \quad (17)$$

where $\mathbf{W}_{10} \in \mathbb{R}^{2 \times d \times d}$, $\mathbf{W}_{11} \in \mathbb{R}^{d \times d}$ and $\mathbf{b}_{10}, \mathbf{b}_{11} \in \mathbb{R}^d$ is learnable parameters, $\mathcal{H} \in \mathbb{R}^d$ is the final output of fusion layer.

3) GCN Decoder: In the TransFOL model, we employ GCN as the decoder component, primarily learning and aggregating local features of drug entities and their neighbouring nodes to accomplish drug-drug interaction prediction tasks.

Specifically, building upon the previous fusion layer module that learns global features, GCN performs information propagation and aggregation operations on the knowledge graph, capturing intrinsic connections and similarities between each drug node and its neighbouring nodes. These local features include the attribute information of the drug itself and encompass relevant knowledge about other biomedical entities (such as diseases and targets) connected to it.

In addition to extracting local features, each layer of GCN recursively combines the information of the current node with its one-hop or multi-hop neighbourhood, forming a richer and more comprehensive node representation. This multi-level neighbourhood fusion enables the model to understand and explore potential drug interaction relationships at different levels.

After obtaining high-level semantic representations of drugs and relationships through Cross-Transformer encoding, the GCN module collects and integrates this contextual information to infer the final drug-drug interaction category or strength, which enhances the model's accuracy and generalization ability in predicting DDIs in complex scenarios.

The aggregation method of a single-layer graph convolution is shown as follows:

$$\mathcal{H}_i^{(k+1)} = \text{ReLU} \left(\sum_{\tau} \sum_{j \in \mathcal{N}_{\tau}^i} \mathbf{W}_{\tau}^{(k)} \mathcal{H}_j^{(k)} \right) \quad (18)$$

where $\mathcal{H}_i^{k+1} \in \mathbb{R}^{d^{(k+1)}}$ represents the hidden state of node i in the k -th layer of GCN, and $d^{(k)}$ indicates the dimension of the node embedded in the k -th layer, and $\mathbf{W}_{\tau}^{(k)}$ is the weight of the edge connecting the k -th layer with node i . $\mathcal{N}_{\tau}^i$ represents the set of direct neighbors of node i , including node i itself.

E. Training and Inference

In this study, we utilize labelled supervision information to train the model. Specifically, we train the model using the inference dataset, which is masked predictions. The dataset contains a set of possible answers to a masked query graph. Given a set of feasible answers $\mathcal{E}_A$ and the trainable input embedding $\mathbf{x}$ for a masked query graph, the loss function per query is formulated as:

$$-\frac{1}{|\mathcal{E}_A|} \sum_{e \in \mathcal{E}_A} \log \frac{\exp(\mathbf{x}^\top \mathbf{u}_e)}{\exp(\mathbf{x}^\top \mathbf{u}_e) + \sum_{e \notin \mathcal{E}_A} \exp(\mathbf{x}^\top \mathbf{u}_e)} \quad (19)$$

where $\mathbf{u}_e$ refers to entity e 's embedding in the decoder. When calculating the loss of one answer, the predictions of other answers are ignored in the loss function.

During training, the model is learned by minimizing the total loss using a stochastic gradient optimizer, such as AdamW [17]. This approach enables the model to optimize its parameters based on the labelled supervision information.

A subgraph of unknown node pairs (u, v) in the knowledge graph is extracted during inference. To predict the relation between u and v , we use the same steps as in the training process to compute the relation vector. The relation with the highest probability is selected as the predicted relation for multi-classification tasks. In contrast, all scores of positive and negative counterparts

TABLE I
SPECIFIC LOGICAL QUERY PROCESSES OF FOUR TASKS

Task	Logical Query Process
1	$\{((drug_1, drug_2))\} \rightarrow \{interaction\}$
2	$\{((drug_1, drug_2))\} \rightarrow \{interactions\}$
3	$\{((drug_1, r_1), (drug_2, r_2)) r_1, r_2 \in doses\} \rightarrow \{response\}$
4	train : $\{(MP_i, r_{m1}, r_{m2})\} \rightarrow \{HP_j\}, \{(disease, r_{d2})\} \rightarrow \{HP_j\}$ test : $\{((MP_i, r_1, r_2), (disease, r_3))\} \rightarrow \{HP_j\}$

of all relations are collected for multi-label tasks to predict the relation. This approach allows the model to generalize to new instances and make accurate predictions.

IV. EXPERIMENTS

This section presents a comprehensive evaluation of the logical query ability of TransFOL in four drug reasoning tasks. To evaluate the model's performance, we conduct four tasks using four different datasets described in Table I.

A. Task Description

This section introduces the four tasks in detail. The logical query process of the four tasks is shown in Table I.

In Task 1 and Task 2, we focus on common DDI tasks. Task 1 evaluates the model's ability to predict the only DDI type between one drug pair. Task 2 evaluates the model's ability to predict multiple types of co-occurring DDI.

Different drug combinations at different doses can affect the target cells differently. Therefore, in Task 3, we conduct multi-dose drug combination-target response prediction. In this task, the predicted anchor node is not limited to a simple drug pair but consists of a pair of drugs at different doses. Such predictions with attribute information are more reliable and realistic, which is challenging for existing prediction methods, demonstrating that TransFOL has more vital drug knowledge discovery ability and better capacity to capture rich knowledge information than the existing DDI models. In Task 3, we assess the model's ability to predict drug on-target effects across multiple drug combinations at different doses, which is critical to shortening the time for drug development.

Furthermore, in Task 4, we simulate a clinical-stage for a drug study experiment, extrapolating human clinical phenotypes (HP) based solely on disease information and mammalian phenotypes (MP). Our method learns the Phenomebrowser heterogeneous graph and only inputs disease and MP information. It should be pointed out that the final predicted HP information has no direct relationship with MP information in the dataset, and the commonly associated drugs only determine their relationship. Therefore, the model can only rely on its understanding of heterogeneous graphs and generalization to make predictions during testing.

The objective of our study is to predict human clinical responses to an unknown drug based solely on mammal responses and corresponding disease information. During the testing phase, the model can predict human symptom data when the entity pair (*mammal symptoms, disease information*) is input.

The training set does not include entity-relationship pairs to evaluate the model's generalization and transferability.

Task 4 includes drug knowledge discovery tasks such as DPI (drug-protein interaction), DTI (drug target interaction), and DHI (drug-human interaction). Existing drug knowledge discovery models cannot solve such complex tasks, highlighting TransFOL's unique advantages. Existing drug knowledge discovery models cannot solve such complex tasks. At the same time, TransFOL can fit different types of entity relationships through query transformations, enabling semantic learning, which fully highlights TransFOL's advantages.

B. Baselines

We compare our model with several baselines. All baseline parameter settings are the optimal parameters of the original text. Since existing DDI models have difficulty handling Task 3 and Task 4, only the logical query baseline is selected for comparison.

- GQE [18] utilizes deep set for intersection (conjunctive) logical operation as learned geometric functions in this space.
- GraIL [19] is for inductive relation prediction on KGs, which uses a local subgraph.
- KGNN [10] adopts GCN to selectively aggregate the neighbour information with high-level layer information for learning the node representation in the KG.
- Q2B [20] embed logical queries and knowledge graph entities into geometric vector spaces as box shapes.
- LogicE [21] utilizes complex query embedding and uses Skolemisation to eliminate existential variables for efficient querying.
- ConE [22] is a query embedding(QE) method based on geometry that represents entities and queries as Cartesian products of two-dimensional cones and can handle all FOL operations.
- MUFFIN [23] adopts a bi-level cross strategy that includes cross-level and scalar-level components to fuse multi-modal features well.
- SumGNN [24] adopts a subgraph extraction module and a self-attention-based subgraph summarization scheme to predict DDI.

C. Experiment Settings

1) **Datasets:** In this study, we use four different datasets to perform four tasks.

- For the multi-class DDI prediction task, we use the Drug-Bank dataset [25], which consists of 1709 drugs(nodes) and 136,351 drug pairs(edges), with 86 pharmacological relationships. Each drug pair can contain one or two relationships. Since more than 99.8% of edges have only one edge type [26], we filter out edges with more than one type in our study.
- For the multi-label DDI prediction task, we use the TWO-SIDES dataset [27], which comprises 645 drugs(nodes) and 46,221 drug-drug pairs(edges), with 200 different

drug side effect types as labels. Each edge may be associated with multiple labels. Following Nováček and Mohamed [11], we retain 200 common DDI types ranging from Top-600 to Top-800 to ensure at least 900 drug combinations per DDI type.

- iii) For the multi-dose drug combination-target response prediction task, we use the DrugCombDB dataset [28], which is based on SPARQL queries and currently contains more than 4 million phenotypic associations of genes, diseases, drugs, pathogens, and chemical entities (metabolites) and 6,055,926 experiment data point, which contains quantitative drug dose-response and drug concentration information of cancer cells, covering 2,887 anti-cancer drugs and 104 types of cancer cells. We extract a subgraph dataset containing the relationship “*drug-dose-target-response*” from it. This sub-dataset lies between traditional DDI datasets and large-scale biomedical datasets, integrating complex interaction information between different dosage conditions of drugs and their biological effects on targets. In terms of community usage, this sub-dataset is suitable for predicting drug combination-target responses at multiple dosages and supports various drug discovery and pharmacological research tasks. These tasks include investigating the impact of drug dosage variation on drug efficacy and side effects, optimizing drug combination therapy regimens, and deciphering drug mechanisms of action. Additionally, researchers can utilize this dataset for secondary analysis, combining machine learning methods to uncover potential dose-response patterns, thereby facilitating the reduction of drug development cycles and improving treatment efficacy.
- iv) For the simulated clinical trial task, we use the Phenomebrowser dataset [29], which contains interactive information extracted from multiple data sources, including a total of 61,146 medical entities (nodes), 1,246,726 medical entity pairs (edges), seven types of node types, and nine semantic relationships. Each type of biomedical concept (*i.e.*, *drug*, *gene*, *disease*, and *phenotype*) is considered a node type, and each type of interaction (*i.e.*, *drug-target-gene*, *drug-treatment-disease*) is considered an edge type. We extract sub-datasets that contain the “*mammal symptom-human symptom-drug-disease*” relationship from it. The sub-dataset exhibits various structured connections, such as the molecular mechanisms of drug-target interactions (*drug-target-gene*) and the therapeutic effects of drugs in practical treatments (*drug-treatment-disease*), among other complex relationships. It aims to simulate symptom responses and drug efficacy evaluations in a clinical setting, providing a unique perspective for cross-species correlation analysis. This sub-dataset is not only suitable for simulating clinical trial prediction tasks but also applicable for exploring the evolution paths of symptoms, validating the effectiveness and safety of drug indications, uncovering potential biological mechanisms of drug effects, and revealing the

commonalities and differences of diseases through comparing symptom manifestations across different species in biomedical research scenarios.

2) Evaluation Strategy: We split the four datasets into training, validation, and test sets in a ratio of 7:1:2. To ensure that the highly imbalanced label distribution does not affect the model’s performance, we make sure that there was no overlap between the train/validation/test sets for all datasets, maintaining their independence.

3) Metrics: We use different metrics to measure the performance of TransFOL in four different tasks. In addition, we also designed a metric $SUB = |Prediction - Truth|$ to measure the average difference between the predicted and actual values in Task 3. When the data accuracy is high, the metric SUB can more truly evaluate the situation in which the predicted results are close but only partially consistent.

4) Parameter Settings: We represent KG entities, relations, and drug structure embeddings as 1024-dimensional vectors. The AdamW optimizer [17] is used to optimize the training loss. The experimental learning rate is set to 0.0001. The model is trained for 500 epochs. The hyper-parameter settings of baselines are the same as reported in their original papers. In Task 1 and Task 2 without additional information, the number of entity encoder layers and relational encoder layers in the separation layer is 2 and 1, respectively; while in Task 3 and Task 4 with complex queries, both are two layers, which are determined by the number of relation types in the task dataset. The number of encoder layers of the fusion layer is eight layers. The hidden side of the last fully connected layers is set to 2048. The parameter sensitivity experiment finds that in Task 1 and Task 2, the best number of GCN decoder layers is 3, and in Task 3 and Task 4, the best number is 1.

D. Main Results

1) Multi-Class DDI Prediction: We present the results of our model and all baselines on the multi-class DDI prediction task on the Drugbank dataset [25] in Table II. Our model outperforms all baselines in predicting DDI. Specifically, our model achieves 0.89% (Accuracy), 0.13% (F1-Score), and 5.46% (Cohen’s Kappa) improvements, respectively, compared to the best baseline.

2) Multi-Label DDI Prediction: We present the results of our model and all baselines on the multi-label DDI prediction task on the TWOSIDES dataset [29] in Table II. TransFOL performs best in the multi-label task. Specifically, it achieves 0.01% (AUC), 0.91% (AUPR), and 1.54% (AP@50) improvements, respectively, compared to the best baseline.

3) Multi-Dose Drug Combination-Target Response Prediction: We present the performance of our model and all baselines on the multi-dose drug combination-response prediction task using the DrugCombDB dataset [28] in Table III. The results indicate that TransFOL achieves the best performance in complex settings. Specifically, our model achieves 0.97% (Hits@1), 3.03% (Hits@3), 1.24% (Hits@10), 0.35% (F1-Score) and

TABLE II
RESULTS OF THE MULTI-CLASS AND MULTI-LABEL DDI PREDICTION TASKS

Methods	Multi-class			Multi-label		
	Accuracy	F1-Score	Cohen's Kappa	AUC	AUPR	AP@50
GraIL [19]	0.8989	0.8131	0.8989	0.9289	0.9110	0.8621
KGNN [10]	0.9089	0.8291	0.8964	0.9075	0.8816	0.8605
MUFFIN [23]	0.9648	0.9495	/	0.9160	0.7033	/
SumGNN [24]	0.9266	0.8685	0.9072	0.9486	0.9335	0.8675
TransFOL	0.9737	0.9508	0.9618	0.9487	0.9426	0.8829

The bold values indicate optimal methods and methods with underlined values indicate suboptimal methods.

TABLE III
RESULTS OF THE MULTI-DOSE DRUG COMBINATION-RESPONSE PREDICTION TASK

Methods	Hits@1	Hits@3	Hits@10	F1-Score	SUB
GQE [18]	0.7663	0.8593	0.9261	0.6949	9.7440
Q2B [20]	0.7399	0.8310	0.9017	0.7002	12.5288
LogicE [21]	0.7507	0.8787	0.9652	0.6943	9.5574
ConE [22]	0.7466	0.8324	0.9277	0.7066	11.2375
TransFOL	0.7760	0.9090	0.9776	0.7101	9.0998

The bold values indicate optimal methods and methods with underlined values indicate suboptimal methods.

TABLE IV
RESULTS OF THE SIMULATED CLINICAL TRIAL

Methods	Hits@1	Hits@3	Hits@10	F1-Score
GQE [18]	0.1518	0.2867	0.3386	0.5528
Q2B [20]	0.1727	0.2733	0.4360	0.5257
LogicE [21]	0.1727	0.2733	0.4360	0.4833
ConE [22]	0.1521	0.2780	0.4240	0.5444
TransFOL	0.1881	0.3027	0.4744	0.5554

The bold values indicate optimal methods and methods with underlined values indicate suboptimal methods.

0.4576%(SUB) improvements, respectively, compared to the best baseline.

4) *Simulated Clinical Trial*: Table IV displays the performance of our model and all baselines in the simulated clinical trial, using the Phenomebrowser dataset [27]. The results demonstrate that our model, TransFOL, achieves the highest performance. Specifically, our model achieves 1.54% (Hits@1), 1.60% (Hits@3), 3.84% (Hits@10), and 0.26%(F1-Score) improvements, respectively, compared to the best baseline.

E. Ablation Study

Enhancement module & GCN: We conduct ablation studies on four experiments to analyze the functions of enhancement modules and GCNs. Table V reports the average results of various model structure predictions in all tasks. Without the enhancement module, the model's performance metrics on the four datasets significantly decreased by 0.64%(Accuracy), 1.04%(AUC), 0.18%(Accuracy), and 0.73%(Hits@1), respectively. Similarly, without the GCN decoder, the model's performance metrics on the four datasets decreased by about 1.15%(Accuracy), 6.67%(AUC), 0.87%(Accuracy), and 0.92%(Hits@1), respectively. Furthermore, without the enhancement module and the GCN decoder, the model's performance metrics on the four datasets drop even more,

demonstrating that both the enhancement module and GCN can improve the model's capacity for complex query reasoning. It should be pointed out that the separation and fusion layers are the basis of the model, so we did not conduct ablation experiments on them.

Hyperparameters: We conduct ablation experiments on the number of GCN layers and report the results of each experiment in Fig. 2. In Task 1 and Task 2, the model's performance initially increases and decreases as the number of GCN layers increases, suggesting that having sufficient and appropriate model depth is crucial for the success of TransFOL. This relationship between depth and receptive field size has been noted in prior research, showing that excessively deep GNNs may suffer from over-smoothing and training instability, leading to degraded performance [30]. However, in Task 3 and Task 4, the model's performance gradually decreases as the number of GCN network layers increases. Additionally, the ablation experiment indicates that the model's performance is worse without the GCN decoder. We think this may be due to the relatively small size of the datasets for Task 3 and Task 4, making them more susceptible to overfitting [31].

F. Case Study

This section presents case studies to demonstrate TransFOL's predictive capability.

Firstly, in the multi-class DDI task, we conduct a detailed analysis of TransFOL's prediction results. Following Yu et al. [24], we focus on the top 10 drug pairs of Quinidine in Table VI. We apply the Drug Interactions Checker tool PubMed, DrugBank, and Drugs.com provided to validate the evidence for these ten drug pairs. Table VI shows that nine out of 10 DDI pairs have supporting evidence, further demonstrating the effectiveness of TransFOL. For instance, our model predicted that Eliglustat would reduce metabolism when combined with Quinidine, which was reported in the literature [32]. Additionally, the DDI pair (*Lisinopril*, *Quinidine*) predicted by the model was not included in the dataset. It may be a potential DDI pair, as Shulyak et al. [33] have conducted some research on this, confirming the association between Lisinopril and Quinidine.

Secondly, in the multi-label DDI task, we provide a case study of the novel drug pair loratadine and ondansetron predicted by the model. Loratadine is an allergy medicine [34], and ondansetron prevents or treats nausea and vomiting after chemotherapy [35]. TransFOL assigns high weights to the three side effect nodes of vision loss, motor retardation, and breast

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON FOUR TASKS

Modules	Multi-class		Multi-label		Multi-dose		Generalized simulated		
	Accuracy	F1-Score	AUC	AUPR	Accuracy	SUB	Hits@1	Hits@3	Hits@10
w/o Enhancement & GCN	0.9563	0.9052	0.8678	0.7939	0.7657	9.2305	0.1713	0.2405	0.3632
w/o Enhancement	0.9673	0.9136	0.9383	0.9285	0.7742	9.1024	0.1808	0.2943	0.4632
w/o GCN	0.9658	0.9105	0.8820	0.8220	0.7673	9.2213	0.1789	0.2538	0.3855
TransFOL	0.9737	0.9508	0.9487	0.9426	0.7760	9.0998	0.1881	0.3027	0.4744

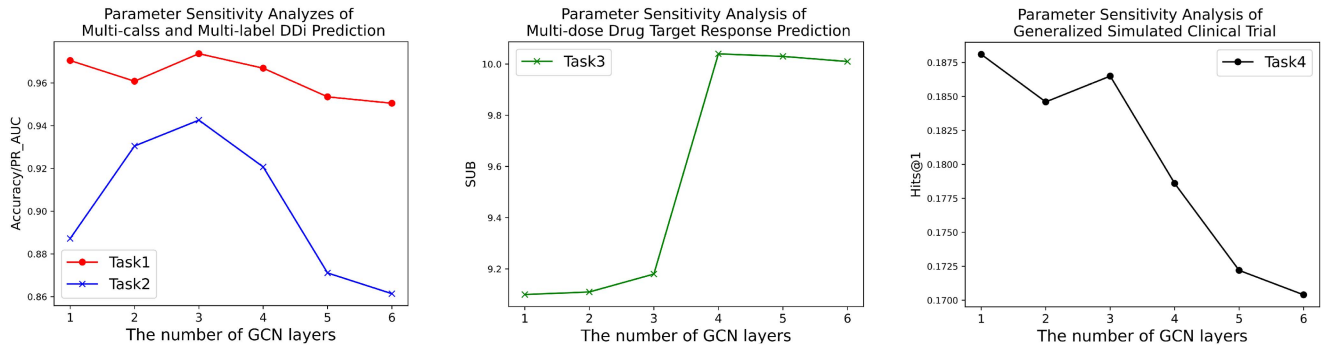


Fig. 2. Parameter sensitivity analysis

TABLE VI
TOP 10 DRUGS THAT MAY INTERACT WITH QUINIDINE

Drug name	Interaction type	Evidence
Eliglustat	The metabolism of Eliglustat can be decreased when combined with Quinidine.	Drug.com
Potassium bicarbonate	Potassium bicarbonate may decrease the excretion rate of Quinidine which could result in a higher serum level.	PMID:5360313 8739822
Sulfasalazine	Sulfasalazine may decrease the excretion rate of Quinidine, which could result in a higher serum level.	DrugBank
Vincristine	Vincristine is a vinca alkaloid used to treat acute leukemia, malignant lymphoma, Hodgkin's disease, acute erythraemia, and acute panmyelosis.	DrugBank
Bretylum	Bretylum may increase the hypotensive activities of Quinidine.	PMID:386786
Potassium	Potassium may decrease the excretion rate of Quinidine which could result in a higher serum level.	PMID:8930193 6025406
Lisinopril	Lisinopril is an ACE inhibitor used to treat hypertension, heart failure, and acute myocardial infarction.	Unconfirmed
Iothalamic acid	Iothalamic acid may decrease the excretion rate of Quinidine which could result in a higher serum level.	DrugBank
Romidepsin	The risk or severity of QTc prolongation can be increased when Romidepsin is combined with Quinidine.	PMID:32449208
Cyclosporine	Cyclosporine is a steroid-sparing immunosuppressant used in organ and bone marrow transplants and inflammatory conditions.	DrugBank

disease. Vision loss is a decreased ability to perceive visual stimuli; motor retardation is a medical condition in which the patient experiences some limitations in basic reflexes and movements. In addition, the combination of loratadine and ondansetron can cause some non-neoplastic or neoplastic conditions affecting the breast. Notably, vision loss is closely associated with motor retardation, a central nervous system problem triggered by a depressed mood. As both drugs risk side effects, our model predicts that there may be central nervous system (CNS) deterioration when taken together, which was supported in the literature [36].

Thirdly, in the multi-dose drug combination-target response prediction task, we also provide a case study of different doses

of combinations predicted by the model against 5-FU and ABT-888. 5-FU is a thymidylate synthase inhibitor widely used in cancer treatment [37], while ABT-888 (Veliparib) is a PARP inhibitor [38]. As shown in Fig. 3, according to the model prediction, when the dose of 5-FU is 0 μ M, the response value of cancer cells is consistently high (the lower the value, the better). However, when the dose of 5-FU is 10 μ M, the response value of cancer cells decreases with the increase of the ABT-888 dose. The overall level is low, meaning ABT-888 alone has almost no anti-cancer effect. After a certain dose of 5-FU, the anti-cancer effect increases with the addition of ABT-888. This result has been reported in the literature [39].

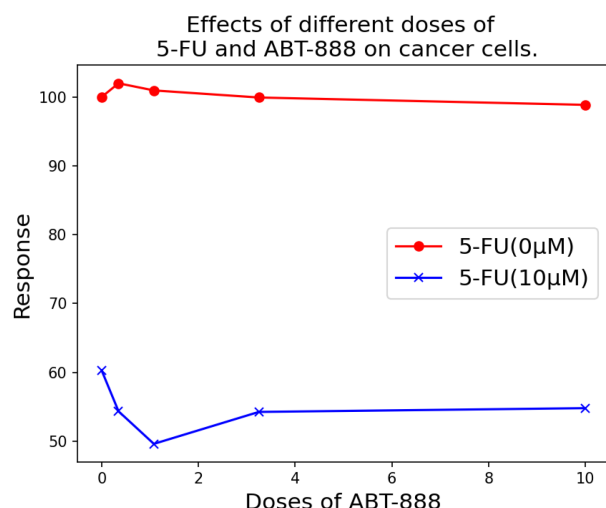


Fig. 3. Responses of different doses of 5-FU and ABT-888 on cancer cells. **Note:** Response refers to the cancer cell activity; the lower value represents better inhibitory effects of the drugs.

Finally, in a simulated clinical trial, we provide a case study of a new drug clinical trial predicted by the model. It is reported that a particular new drug developed for impetigo can cause dysregulation of physiological homeostasis in mammalian experimental models [40]. According to the model prediction, adverse reactions such as skin photosensitivity and cheilitis may occur in human clinical trials, with skin photosensitivity assigned the highest weight, which has been reported in the literature [39], [41] and further demonstrates the effectiveness of TransFOL and proves that TransFOL can help to reduce the clinical trial cycle of drug development.

G. Limitation

Although TransFOL has shown good reasoning ability in complex settings, there are some limitations:

Error Accumulation: Biomedical datasets are sparse and contain significant noise, which can lead to TransFOL learning erroneous knowledge and propagating those errors as the number of hops increases during inference. For instance, if the input training data contains the incorrect information that “Drug A interacts with Drug B,” during the inference stage, our model may further infer that “Drug C, which belongs to the same category as Drug A and has similar neighbourhood relationships with A, interacts with Drug B,” and then infer that “Drug D, similar to Drug B, interacts with Drug C”... However, these DDI relationships do not exist. In future research, we will introduce techniques such as type-based negative sampling strategies and adversarial learning to help the model better correct errors.

Query Templateization: Although TransFOL can perform various complex predictions, it requires preprocessing the data into corresponding fixed formats using query templates. This templateized query approach limits the model’s applicability. For example, additional manual intervention and data preprocessing are often needed to meet fixed query templates, increasing the cost of transforming raw data into a format acceptable to the

model and potentially losing useful information. Additionally, as the knowledge base grows and updates, discoveries and data may not align with existing query templates, making it difficult for the model to quickly absorb and utilize this new knowledge for real-time updates and predictions. In future research, we will explore graph-structured queries as an alternative to fixed data templates and dynamically generate adaptive queries by learning the features and patterns of the data.

The key advantage of graph-structured querying is its ability to represent and query drug-drug interactions more flexibly. Compared to fixed data templates, graph-structured querying can dynamically generate query statements based on specific data characteristics and patterns, better accommodating different scenarios. However, while graph-structured querying is theoretically appealing and potentially beneficial, we did not directly adopt this approach in the current study, as extended research is needed to adequately address the challenges of using graph-structured querying. Employing graph-structured querying may require redesigning and retraining the model to better adapt to the new querying method, which could entail more complex model architectures and algorithms and additional data and computational resources to support model training and inference.

V. CONCLUSION

This paper presents a novel approach called TransFOL for complex DDI prediction. We build upon a logic query formulation and leverage a Cross-Transformer module to encode the intricate semantic information between entities and a GCN decoder for further decoding and integration. Our experiments on biomedical datasets demonstrate that TransFOL achieves excellent performance across four benchmark datasets spanning multiple tasks, outperforming several strong baselines and showcasing its powerful drug knowledge discovery capabilities and strong generalization ability.

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